Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
Course no: CSE-2202

Lab 1

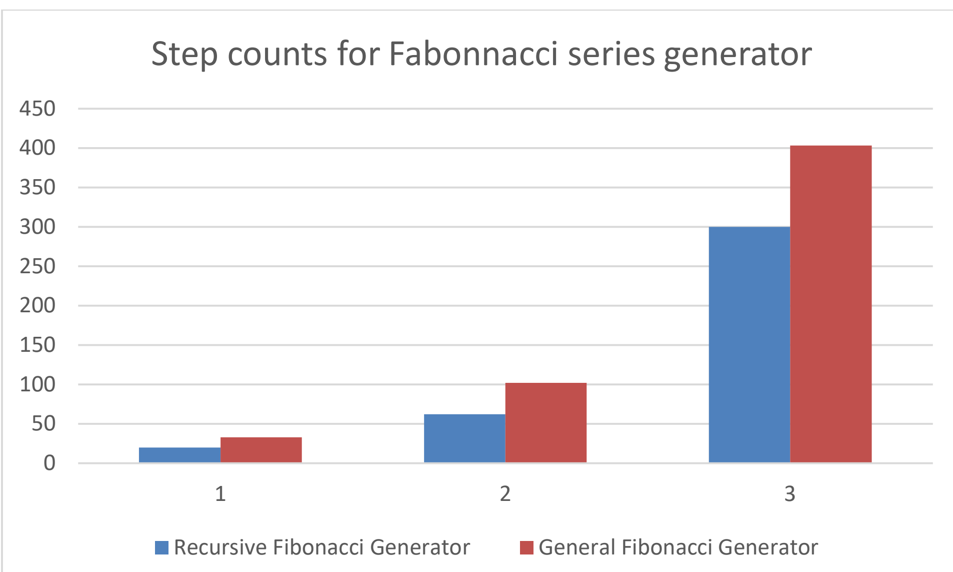
Task-1: Generate a recursive algorithm for creating the series of Fibonacci numbers up to a given number of elements. The number of elements should be read from a file **fibonacci.txt**. Your task is to implement the code of your algorithm and then find out the total number of steps count. Finally compare the number of steps (i.e., at least 10 comparisons) with an algorithm which do not use recursive method for generating the Fibonacci numbers. Present the comparisons in by a graphical means and describe accordingly.

Example

Algorithm name	N1	N2	N3
Recursive Fibonacci Generator	20	62	300
General Fibonacci Generator	33	102	403

The above values are simply descriptive purposes and it is not the exact number of step counts.

Graphical Presentation:



Your report should contain the followings:

1. Introduction/ Problem statements
2. Details description and algorithms
3. Implemented Code
4. Sample Input and Output (Include the graphs if required)
5. Discussion and conclusion